Solution.

BA865 – SPRING 2026 (V1)

QUIZ 2

1. **Convolution Operation (2 pt).** What is the first value in the output feature map here? That is, demonstrate the first calculation of the convolution operation, applying the filter to the first position of the input tensor/matrix.

Input Tensor:

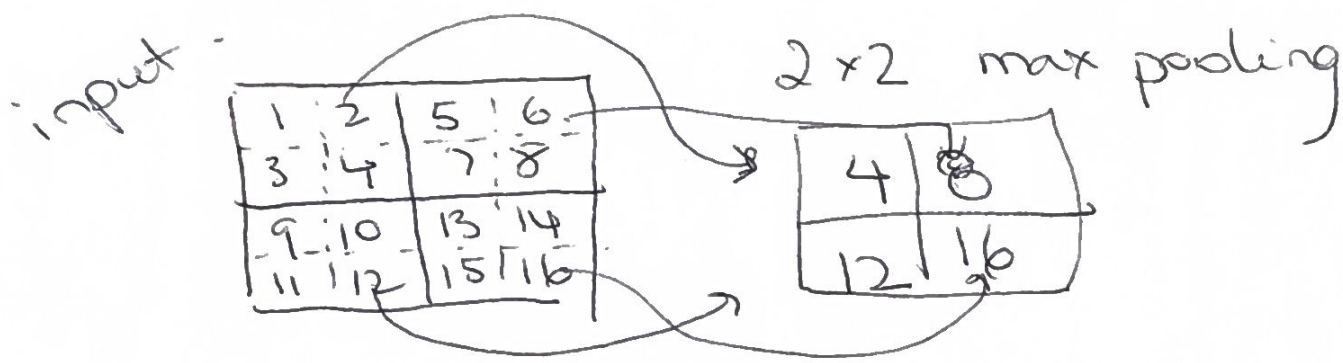
1	2	3	0	1
0	1	2	1	0
1	0	1	2	1
2	1	0	1	0
0	1	2	0	1

$(1\times1)+(2\times0)+(3\times-1)$
$+(0\times1)+(1\times0)+(2\times-1)$
$+(1\times1)+(0\times0)+(1\times-1) = 1 - 3 - 2 + 1 = -4$

Filter:

1	0	-1
1	0	-1
1	0	-1

2. **Pooling (1 pt).** Show (draw for me) what a max pooling layer does to an input tensor?



3. **Filter Count / Feature Map Size (2 pts).** How does the number of filters typically change as one goes deeper in a CNN, and how do feature map sizes change? Why does this happen?

→ Filters increase as one goes deeper in the network, but the size of their outputs (feature maps) gets smaller.

→ Size declines due to pooling or lack of padding in convolutions.

→ Filter count is increased to accommodate the increased size/scope of the feature space, e.g. line detectors vs. concept detectors - e.g., dog, cat, tree, humor, etc.